MIL-P-60015A (PA)
22 May 75
SUPERSEDING
MIL-P-60015 (MU)
30 January 1964

MILITARY SPECIFICATION

PRIMER, PERCUSSION, M60A1
LOADING, ASSEMBLING, AND PACKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of percussion primer designated as the M60A1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-223	-	Powder, Black
MIL-A-2550	-	Ammunition and Special Weapon, General Specification for

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-1168	-	Ammunition Lot Numbering
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

FSC: 1390

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DRAWING

ARMY

8839466	-	Primer, Percussion M60A1 Assembly
8858935	-	Box, Packing, Ammunition, For Primers M28A2, M28B2, or M60A1
8858940	-	Box, Set-up, Packing Ammunition for Primers

(Copies of specifications, standard, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Material and parts shall be in accordance with applicable drawings and specifications.

3.2 Primer Assembly.-The primer assembly shall comply with all requirements specified on drawing (dwg.) 8839466 and with all requirements specified in applicable specifications.

3.3 Moisture Content of Black Powder -The moisture content of black powder at the loading station at the time of loading, shall not exceed 0.5 percent.

3.4 Head Assembly

3.4.1 Non-functioning.-The head assembly shall not function by an applied energy of 47.52 inch ounces minimum (min.) when tested as specified in 4.3.1.1.

3.4.2 Functioning.-The head assembly shall function by an applied energy of 159.20 inch ounces maximum (max.) when tested as specified in 4.3.1.2.

3.5 Primer Assembly

3.5.1 Functioning.-The primer assembly shall function without audible hangfire or evidence of ruptured, broken or detached parts when tested as specified in 4.3.2.

3.5.2 Pressure Resistance.-The primer assemblies shall withstand a pressure of 40,000 plus or minus 4,000 pounds per square inch without evidence of rupture, or any part of the primer becoming detached, when tested as specified in 4.3.2.

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3.6 Workmanship.-All parts shall be dry and free of cracks, chips, dirt, grease, rust and other foreign material. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. In addition, the Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.2 Inspection Provisions

4.2.1 Inspection Lot Formation.-The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified in MIL-STD-105. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot shall contain:

a. Metal parts of one lot interfix number from one manufacturer.

b. Black powder from not more than one lot from one manufacturer.

from 4.4.1.2) shall be randomly selected from each lot and tested

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c. Primer (dwg. 7645339) herein after referred to as primer element, of one lot interfix number from one manufacturer.

4.2.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Sampling plans shall be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Body (see Dwg. 8838089 covering a detail of dwg 8839466).

Categories	Defects	Method of Inspection
Critical:		
	1. Flash hole missing.....	Gage
	2. Crack or split	Visual
Major: None defined		
Minor:		
	201. Evidence of poor workmanship (see 3.6).....	Visual

4.2.2.2 Head assembly (see dwg. 8839466).

Categories	Defects	Method of Inspection
Critical:		
	1. Percussion element above flush.....	Visual-Straight-edge
Major:		
	101. Percussion element missing, cracked or inverted.....	Visual
	102. Flash hole in head completely obstructed	Visual

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103. Percussion element loose .....Manual
104. Crimp of percussion element to head missing or incomplete....Visual

Minor:
201. Seal missing or incomplete.....Visual
202. Evidence of poor workmanship.....Visual

4.2.2.3 Primer assembly, prior to assembling diaphragm (see dwg. 8839466).

Categories	Defects	Method of Inspection
Critical:	None defined	
Major:		
	101. Weight of charge	Balance
	102. Liner fails to cover all holes	Visual
	*103. Flash hole not completely sealed	Visual
	104. Black powder missing	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.6)	Visual

*Sketches 1, 2, and 3 of Figure 1 indicate an acceptable primer. Also, a primer with holes displaying the conditions in 4, 5, and 6 may be classed acceptable provided that the visible liner is thoroughly impregnated with paint or lacquer and not more than 20 percent of the holes have this condition.

4.2.2.4 Primer assembly (see dwg. 8839466).

Categories	Defects	Method of Inspection
Critical:	None defined	
Major:		
	101. Head assembly loose	Manual
Minor:		
	201. Length max.	Gage
	202. Diaphragm improperly sealed	Visual

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203. Diaphragm above flush.....Visual
204. Crimp of body to head missing or incomplete.....Visual
205. Evidence of poor workmanship (see 3.6) .....Visual

4.2.2.5 Set-up box, prior to sealing (see dwg. 8858940).

Categories	Defects	Method of Inspection
Critical:	None defined	
Major:	None defined	
Minor:		
	201. Any packing material missing....Visual	
	202. Primer missing .....Visual	
	203. Separator damaged to the extent that primers touch each other...Visual	

4.2.2.6 Sealed set-up box (see dwg. 8858940).

Categories	Defects	Method of Inspection
Critical:	None defined	
Major:	None defined	
Minor:		
	201. Sealing strip torn, incomplete badly wrinkled, or otherwise fails to seal carton.....Visual	
	202. Carton torn, cut or punctured...Visual	
	203. Marking missing, misleading or unidentifiable.....Visual	
	204. Contents loose.....Manual	

4.2.2.7 Sealed barrier bag (see dwg. 8858940).

Categories	Defects	Method of Inspection
Critical:	None defined	

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Major:
101. Bag punctured, torn or cut.....Visual
102. Seal mislocated or incomplete...Visual

Minor: None defined

4.2.2.8 Packing box, prior to sealing (see dwg. 8858935).

Categories	Defects	Method of Inspection

Critical: None defined

Major:
101. Carton missing.....Visual

Minor: None defined

4.2.2.9 Packing box, sealed (see dwg. 8858935).

Categories	Defects	Method of Inspection

Critical: None defined

Major:

101. Box broken, board loose or protruding nail.....Visual
102. Strapping missing, broken or loose.....Visual
103. Cover improperly assembled.....Visual

Minor:
201. Marking misleading or unidentifiable.....Visual
202. Strapping improperly engaged or mislocated.....Visual
203. Contents loose.....Visual

4.2.3 Testing

4.2.3.1 Head Assembly

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4.2.3.1.1 Non-functioning (see 3.4) Critical defect - Primers subjected to this test shall be destroyed..

4.2.3.1.1.1 Beginning with the first lot produced, and continuing until three consecutive lots have complied with the acceptance criteria specified, a sample of 150 head assemblies shall be randomly selected from each lot for the test. If any sample fails to comply with requirement the lot shall be rejected. The test shall be performed as specified in 4.3.1.1.

4.2.3.1.1.2 After three consecutive lots have met the criteria of 4.2.3.1.1.1, a sample of 50 head assemblies shall be randomly selected from each lot for this test. If one or more primers fails to comply with the applicable requirement, the lot shall be rejected. The test shall be performed as specified in 4.3.1.1 using equipment in accordance with 4.2.4.

4.2.3.1.2 Functioning (see 3.4) Major defect -

4.2.3.1.2.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified two hundred and fifty samples shall be randomly selected from each lot for this test. If three or more primers fail to comply with the applicable requirements, the lot shall be rejected. If one or two defects are found, a second sample of five hundred primers shall be tested. If the combined number of defectives in both the first and second sample is three or more, the lot shall be rejected. The test shall be performed as specified in 4.3.1.2 using equipment in accordance with 4.2.4.

4.2.3.1.2.2 After three consecutive lots have met the criteria of 4.2.3.1.2.1, a sample of 125 primers shall be randomly selected from each lot for this test. If two or more primers fail to comply with the applicable requirement, the lot shall be rejected. If only one defect is found, a second sample of 250 primers shall be tested. If the combined number of defectives in both the first and second sample is two or more the lot shall be rejected. The test shall be performed as specified in 4.3.1.2 using equipment in accordance with 4.2.4.

4.2.3.2 Primer assembly (see 3.5) Major defect.-Twenty-five samples shall be randomly selected from each lot for this test. If one or more assemblies fail to comply with the requirement, the lot shall be rejected. The test shall be performed as specified in 4.3.2.

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4.2.3.3 Moisture Content of Black Powder (see 3.3) Major Defect.-The contract shall provide adequate controls to assure the compliance of each lot with the applicable requirement. For verification, and just prior to start of each eight hours loading production, the contractor shall test one sample of powder selected from that quantity, or sub-lot of powder that will be used in the eight hour loading production of primer assemblies. Composite samples shall not be used. If the moisture content of the samples exceeds the requirement, and loading has not begun, that sub-lot of powder represented by the sample shall be rejected. If powder from a sub-lot with excessive moisture has been used in loading the remaining unloaded powder and all assemblies containing the non-conforming powder shall be rejected.

4.2.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" headings in 4.2.2 and 4.2.3 (see 6.2).

4.3 Test methods and procedures

4.3.1 Head Assembly

4.3.1.1 Non-functioning.-The loaded heads, prior to assembling to the body, shall be assembled into the test fixture (see 4.2.4). A steel ball weighing a min. of 7.92 ounces shall be dropped from a height to give the required impact force (approximately 6 inches).

4.3.1.2 Functioning.-The test shall be performed as specified in 4.3.1.1 except that the weight of the ball shall be 7.96 ounces max., and the height shall be adjusted to give the required impact force (approximately 20 inches).

4.3.2 Primer assembly.-The primers shall be assembled to the complete round for which the primer was designed and fired from the gun for which the round is standard. The primers shall be observed to determine compliance with the requirement.

4.3.3 Determination of moisture content of black powder.-The moisture content of the black powder shall be determined in accordance with the procedure specified in Specification MIL-P-223.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

5.1.1 Level C.-The primer assemblies shall be preserved and packaged in accordance with dwg. 8858940.

5.2 Packing

5.2.1 Level C.-The cartons shall be packed in accordance with dwg. 8858935.

5.3 Marking.-Marking shall be in accordance with dwg. 8858940 and 8858935.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" heading Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.2.2 and 6.2.3 should he elect to do so.

6.2.1 SIE Designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

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c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.2.2 Contractor Designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings of other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.2.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204 "Inspection Equipment Design" for guidance. The item detail specification number, paragraph number, and defect number from Section 4 shall be referenced on each contractor design together with the component or assembly drawing number, revision letter and date to which the specific design applies.

6.2.3 Submission of Designs for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements Lists (DD Form 1423). Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Designs shall be submitted for approval to Commander, Picatinny Arsenal, ATTN: SARPA-QA-T-T, Dover, New Jersey 07801.

6.3 Check test for possible deterioration of percussion artillery primers.-The following tests are designed to prevent the loading of artillery primers which have deteriorated into a major ammunition item. The test shall be performed on artillery primers from each lot which has been stored under normal conditions for more than two years or under adverse

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conditions for any period of time. If the specification for the ammunition item into which the artillery primers are loaded does not require that the check test be performed, the contracting officer should insert a provision for the performance of the check tests in the contract.

6.3.1 Head Assembly Functioning (see 3.4.2).-The sampling plan for the test shall be in accordance with Table II. Any head assembly which fails to comply with the applicable requirement shall be classed defective. The test shall be performed as specified in 4.3.1.2, except that the flash tube and the head assembly shall be separated in a manner that will not damage the head assembly to the extent that the validity of the test results would be questionable.

TABLE II

Sample	Sample Size	Size	Combined Samples Acceptance Number	Combined Samples Rejection Number
First	75	75	*	2
Second	75	150	0	3
Third	75	225	0	3
Fourth	75	300	1	4
Fifth	75	375	2	5
Sixth	75	450	2	5
Seventh	75	525	4	5

*No acceptance on this sample.

6.3.2 Moisture content of black powder.-A sample of 25 primers shall be selected for this test. The black powder shall be removed from the flash tube by any suitable means. The moisture content of the powder shall be determined in accordance with the method specified in Specification MIL-P-223. If the moisture content of any of the powder tested exceeds 0.7 percent, the lot of primers shall be rejected.

Since black powder is hygroscopic, it is advisable to place the powder in a dry, air-tight container immediately upon removal, and to run the determination as soon as possible thereafter.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1390-A104

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FIGURE 1

ARTILLERY PRIMERS
STANDARDS FOR PAINT CONDITIONS IN FLASH HOLES

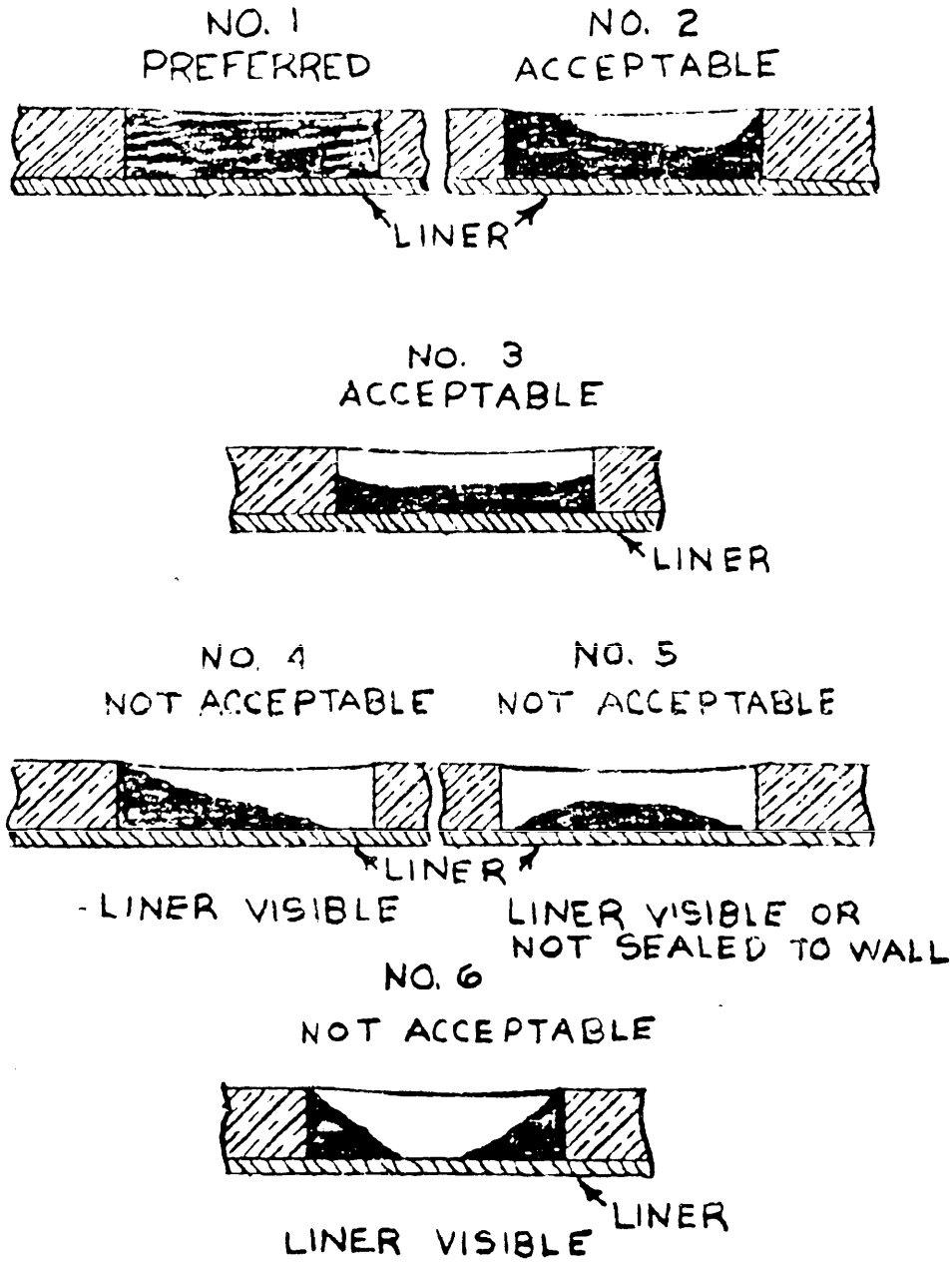


FIGURE 1

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DEPARTMENT OF THE ARMY
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COMMANDER
Picatinny Arsenal
ATTN: SARPA-QA-A
Dover, New Jersey 07801

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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